Machine-learning emulation of precipitation $\delta^{18}O$ reveals its cross-model transferability across PMIP4 palaeoclimate simulations

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Abstract

Precipitation $\delta^{18}O$ links climate models to a wide range of palaeoclimate proxies, yet most models in the Palaeoclimate Modelling Intercomparison Project do not carry water isotopes and cannot be compared with isotope records directly. An emulator of precipitation $\delta^{18}O$ is trained here on the 8199-year isotope-enabled AWIESM2-wiso transient simulation, with seven regression methods from four learner families relating monthly $\delta^{18}O$ to fourteen standard climate fields. On years withheld from training the emulator recovers the interannual variability of the precipitation-weighted annual $\delta^{18}O$ at a land-mean temporal correlation of 0.75. The correlation remains at 0.66 when the emulator is transferred to MPI-ESM-wiso, an independent isotope-enabled model, and the mid-Holocene minus pre-industrial change of that model is reproduced at a spatial correlation of 0.87. Internal fit turns out to be a poor guide to transfer, because kernel and neural learners match the tree ensembles inside the training model and then lose most of their skill on a foreign one. We find that predicting anomalies instead of absolute values restores all seven methods to correlations between 0.85 and 0.92, which identifies the removal of the inter-model base state, rather than the learner family, as the main control on transferability. Applied to nine PMIP4 mid-Holocene simulations the emulator produces a pronounced depletion over the Sahara and the Sahel that at least seven of the nine models share, the isotopic expression of the Green Sahara. Changing the climate model moves the reconstructed change three times as much as changing the learning method. The emulated fields reproduce the spatial temperature effect and its latitude dependence, while the temporal slopes between the two climate states reach only about half of the spatial ones, which cautions against calibrating a palaeothermometer on modern gradients alone.

1 Introduction

Water molecules carry oxygen in more than one isotopic form, and the heavier form condenses more readily than the lighter one. Every phase change along an air mass trajectory therefore leaves a mark, and the composition that finally reaches the ground, expressed as $\delta^{18}O$, integrates the whole history of evaporation, transport and rain-out (Dansgaard, 1964). That integration is what makes the quantity valuable: a single number carries information about temperature, moisture source and circulation at once.

It also makes the quantity difficult to read. Speleothems, ice cores, lake and marine sediments preserve past $\delta^{18}O$, and those archives are the backbone of what is known about climate before the instrumental era. But an archive gives a point, not a field. Records are clustered where caves and ice exist and are absent over most of the ocean and the low-latitude continents, their temporal resolution ranges from annual layers to samples separated by centuries, and each one records the isotopic signal filtered through its own depositional and diagenetic history. The deeper difficulty is interpretive rather than logistic. Because $δ^{18}O$ responds to several processes at once, a shift measured in a speleothem is compatible with a change in temperature, in rainfall amount, in the season when rain falls, or in where the moisture came from, and the record alone cannot separate them.

Resolving that ambiguity requires a physical model that carries the isotopes themselves. Isotope-enabled general circulation models do exactly this, tracking the heavy and light species separately through evaporation, condensation and transport, and they allow a simulated field to be placed directly beside a measured record (Cauquoin et al., 2019; Shi et al., 2023; Bühler et al., 2021). Such comparisons have been made for the last millennium (Bühler et al., 2022) and for the monsoon regions of the mid-Holocene (Parker et al., 2021), and they are precisely what a model without an isotope scheme cannot enter.

Very few such models exist, and the reason is structural rather than incidental. An isotope scheme is not a module that can be attached to an arbitrary host: the tracers must follow the water through every parameterisation that moves or changes the phase of water, so the implementation has to be built into a particular model and rebuilt for the next one. Isotope development consequently lags the climate models it depends on, and it lags them even within a single modelling centre, where the isotope-enabled branch is typically some versions behind the main development line. Running the result is also expensive, because each isotopic species duplicates part of the water cycle. The arithmetic of the field follows from this. Isotope-enabled models are scarce at the atmospheric level (Bong et al., 2026) and scarcer still where a palaeoclimate application needs them, in a fully coupled Earth system. Worldwide there are six: HadCM3 (Tindall et al., 2009), iLOVECLIM (Roche, 2013; Roche and Caley, 2013; Caley and Roche, 2013), iCESM1 (Brady et al., 2019), MPI-ESM-wiso (Cauquoin et al., 2019), AWI-ESM-wiso (Shi et al., 2023) and, most recently, MIROC6-iso (Li et al., 2026). Set against the several dozen models contributing to CMIP6 and its palaeoclimate component, the imbalance is stark. Most simulations contributed to the Palaeoclimate Modelling Intercomparison Project therefore output temperature, precipitation and circulation, but not $δ^{18}O$, and cannot be placed alongside isotope records without an intermediate step.

Two routes have been used to bridge this gap. The first is to run an isotope-enabled model, with the costs just described. Proxy system models and data assimilation instead translate between climate fields and proxy space through forward operators or statistical priors (Dee et al., 2015; Hakim et al., 2016). A learned emulator offers a third route. If the mapping from climate variables to $δ^{18}O$ can be learned from a model that does carry isotopes, the same mapping can then be applied to the climate fields of a model that does not. Learned emulators of standard climate fields have become a mature tool that reproduces model output at a small fraction of the cost (Tebaldi et al., 2025; Cresswell-Clay et al., 2025), yet water isotopes have remained outside their scope.

The same computational argument has already been made, and acted on, for other expensive tracers. Marine biogeochemistry raises the cost of an ocean physics model by roughly an order of magnitude, and deep-learning emulators of those schemes now reproduce their behaviour from daily to decadal timescales (Skákala et al., 2026), while learned components have been folded into biogeochemical data assimilation (Higgs et al., 2026). Water isotopes present the same structure of problem, an expensive tracer riding on a cheaper physical model, but the corresponding development has not taken place. That the gap is recognised is clear from the recent publication of the first machine-learning-ready benchmark dataset assembled specifically for water isotope emulation (Li et al., 2025), which was motivated by the absence of such a resource.

Machine learning has instead been applied to water isotopes mostly as a spatial interpolation problem. Random-forest and related methods predict station or gridded precipitation $δ^{18}O$ from meteorological and geographic predictors across Europe (Nelson et al., 2021; Erdélyi et al., 2023), Australia (Falster et al., 2026), China (Cui et al., 2025; Wang et al., 2025; Liu et al., 2025), Southeast Asia (Heydarizad et al., 2024), the Middle East (Heydarizad et al., 2023) and the Arctic Ocean (Klein et al., 2024). The same approach has been extended from precipitation to other reservoirs, to surface water across China (Wu et al., 2024), to groundwater (Cemek et al., 2022) and to the Mediterranean Sea (Astray et al., 2021). Tree ensembles recur as the strongest learner in several of these comparisons, which is consistent with what we find here, but all of them share a common design: the predictors are observed or reanalysed fields and the target is an observed isotope value, so the product is an isoscape rather than an emulator of a climate model, and the question of transfer between models does not arise.

Only one study has emulated the isotope field of a general circulation model. It used a convolutional network within a single last-millennium simulation (Wider et al., 2023) and reported that the network skill fell to the level of pixel-wise linear regression when the emulator was applied to a different climate model. The wider emulator literature has meanwhile identified cross-model transfer as a general difficulty, since different models carry different mean-state biases and parameterisations (Tebaldi et al., 2025). Whether an emulator trained on one isotope-enabled model can be transferred to others, and how much skill survives that transfer, has therefore not been quantified for water isotopes.

In the present study we train an emulator of precipitation $\delta^{18}O$ on a long isotope-enabled transient simulation and test its transfer to independent climate models. Seven regression methods drawn from four learner families are compared, the emulator is validated against a second isotope-enabled model that supplies ground truth, and it is then applied to nine PMIP4 mid-Holocene simulations that lack isotopes. The transfer is quantified against a self-reconstruction upper bound. The uncertainty of the emulated product is separated into a contribution from the choice of climate model and a contribution from the choice of learning method, and the emulated fields are finally examined for the classical isotope-climate relations they should contain if the learned mapping is physical.

2 Results

2.1 Skill within the training model

The upper bound of what the emulator can achieve is set by applying it back to the simulation it was trained on. One model year in eight enters the training set, which leaves 7174 of the 8199 transient years unseen, and the skill is evaluated on those years alone. All seven learners recover the interannual variability of the precipitation-weighted annual $\delta^{18}O$ over most of the land surface (Fig. 1, first column). The land-mean temporal correlation reaches 0.75 for gradient boosting and histogram gradient boosting, 0.74 for the random forest and the neural network, 0.71 for the support vector regression, 0.69 for kernel ridge and 0.64 for the linear baseline. Skill is highest over the monsoon domains and the mid-latitude continents and falls in the dry subtropical interiors, where the precipitation weighting rests on few wet months.

Two things about this first column matter for everything that follows. The seven learners are separated by only 0.11 in land-mean correlation, so within the training model the choice of method is close to immaterial. And the anomaly framing changes almost nothing here, moving the same seven from 0.64–0.75 to 0.70–0.74 (Fig. 1, second column), which is what one expects when there is no second model and therefore no base-state difference to remove.

The same behaviour holds for the low-frequency evolution of the transient. Regional precipitation-weighted $\delta^{18}O$ follows the orbital-scale trends of the eight thousand years closely, and the seven methods stay within a narrow band around the truth, with regional correlations between 0.46 and

0.97 and biases within 0.15 ‰ (Fig. 2, and Fig. S1 for the same regions in absolute δ^{18}O). The absolute pre-industrial field is recovered almost exactly, with a land RMSE of 0.28 ‰ for gradient boosting and 0.84 ‰ for the linear baseline (Fig. 7, first column), and the mid-Holocene minus pre-industrial change reaches a spatial correlation of 0.87–0.98 against the truth (Fig. 6, first two columns). This self-reconstruction carries no inter-model offset, so it isolates the part of the error that comes from the learning problem itself. Everything that follows is measured against it.

2.2 Transfer to an independent isotope-enabled model

The emulator is next applied to MPI-ESM-wiso, a model that took no part in training and that simulates water isotopes explicitly, so a full field of truth exists for comparison. Interannual skill falls by about 0.1 in correlation for the tree ensembles, from 0.75 within the training model to 0.64 for the piControl state and 0.66 for the mid-Holocene (Fig. 1, third and fourth columns). That gap is the structural cost of learning δ^{18}O in one model and predicting it in another.

The four non-tree learners do not merely lose that much. They fall to 0.43–0.48, roughly the level of the linear baseline, even though within the training model they scored 0.69–0.74. The ranking of the methods inside the training model therefore does not survive the transfer, and the ordering of the first column bears almost no relation to the ordering of the third. This is the central methodological result, and it is examined next.

The penalty also has a systematic component that a correlation cannot show. Regional time series of the raw reconstruction run parallel to the truth but displaced from it, by as much as 12.2 ‰ for the linear baseline over Africa (Fig. S2, and Fig. S3 for the piControl state). The displacement is a difference in base state between the two models rather than a failure to track variability, which is what motivates the anomaly framing. Drawing the piControl and the mid-Holocene as one continuous series against a single piControl reference makes the point directly: the emulated level tracks the truth within a few tenths of a permil in most regions, and the step between the two states is recovered rather than the absolute level (Fig. 3).

2.3 Learning method and preprocessing framing

Two choices are open to anyone building such an emulator: which learner to fit, and whether to work with absolute fields or with anomalies. Their effects turn out to be very unequal.

Under the raw framing the learner matters a great deal for the mid-Holocene change. The three tree ensembles reach a spatial correlation of 0.84–0.87 against MPI-ESM-wiso, the linear baseline reaches 0.37, the neural network 0.29, the support vector regression 0.11, and kernel ridge collapses to −0.02 (Fig. 6, third column). The collapse is visible as an almost blank map: the Green Sahara depletion that dominates the truth is simply absent.

Under the anomaly framing the same seven learners recover to 0.85–0.92 (Fig. 6, fourth column), and the spread between them nearly closes. Removing the base state of the target model, rather than choosing a different learner, is what makes the reconstruction transferable. The trees remain the most robust family in the sense that they alone work under both framings, but they are no longer the only workable choice.

Where the emulator looks for its predictors explains part of this. Averaged over the whole domain the answer is always surface temperature, which says little, but split by region the tree ensembles use surface temperature across the mid-latitude continents, boundary-layer humidity over the two ice sheets, and precipitation over Africa and Australia (Fig. 4). That is the textbook division between the temperature effect and the amount effect, and it emerges from a per-grid-point regression in which latitude and longitude were deliberately excluded from the predictor set.

The linear and neural-network models select quite different predictors in the same regions, which is consistent with their failure to transfer: a combination that is statistically effective within one model need not be physically grounded.

2.4 Absolute fields and the amplitude error

Correlation is not the whole account, because it is blind to amplitude. Every method reproduces the spatial pattern of the pre-industrial field, with spatial correlations of 0.97 or better and maps that are hard to tell apart by eye (Fig. 7). The error is in the level. Against MPI-ESM-wiso the land RMSE ranges from 1.2 ‰ for the random forest to 6.4 ‰ for the linear baseline, a fivefold spread that the absolute colour scale cannot show, and it is largely a uniform offset rather than noise: the land-mean bias is 0.5 ‰ for the random forest but 4.2 ‰ for the linear baseline and 5.5 ‰ for kernel ridge. The zonal profiles make the same point per latitude, with the self-reconstruction bias near zero everywhere and the cross-model bias swinging by several permil.

This is a different failure mode from the interannual one and it constrains what the product may be used for. Shape is reliable; absolute level is not, unless the method is chosen with that in mind.

2.5 Application to the PMIP4 ensemble

The emulator is then applied where no isotope-enabled model exists. Nine PMIP4 models supply the climate fields for the mid-Holocene and the pre-industrial state, and the emulator returns a $\delta^{18}O$ change for each of them. The mean over the nine models and the seven learners is dominated by a depletion over northern Africa and the Sahel exceeding 2 ‰ at its core, with a second centre over the Tibetan Plateau and central Asia (Fig. S4). At least seven of the nine models agree on the sign of the change over 43 % of the land surface, including the whole Saharan and Sahelian centre. The African depletion is the isotopic expression of the strengthened West African monsoon and the wetter Sahara of the mid-Holocene (Tierney et al., 2017; Brierley et al., 2020), and weak enrichment appears over South America, Australia and parts of the southern high latitudes.

The ensemble also separates the two sources of uncertainty that an emulated product carries. Changing the climate model shifts the reconstructed change by 0.20 ‰ in the land mean, while changing the learner shifts it by 0.07 ‰. The choice of climate model therefore matters about three times more than the choice of learner, and the uncertainty of the product is dominated by the models it is applied to rather than by the machine learning. This holds only under the anomaly framing. Under the raw framing the apparent spread across learners is inflated to 0.28 ‰ by the same kernel and linear collapse seen above, which would misattribute a failure of two methods to genuine methodological uncertainty.

2.6 Comparison with the geological record

Every comparison so far has been between models. The final test replaces the reference with measurements: 63 GNIP stations with records longer than four years, 51 speleothem entities from SISALv2 and 15 ice cores, matched to the model grid by bilinear interpolation (Fig. 8).

The emulated pre-industrial field tracks the observations about as well as the physical model does. Against the GNIP stations the emulator reaches a correlation of 0.82 with an RMSE of 2.6 ‰, against MPI-ESM-wiso's 0.86 and 2.0 ‰; against the speleothems both reach 0.76. Applying the emulator to PMIP4 predictors instead of MPI predictors costs little, 0.80 against the stations and 0.74 against the speleothems, which is the case that matters because those models have no isotopes of their own. Statistics are reported by archive rather than pooled, because a single correlation over archives spanning -57 to -1 ‰ would mostly measure the offset between polar ice and tropical caves.

The mid-Holocene change is the harder and more relevant test, and here the comparison is closer than the model-to-model figures would suggest (Fig. S5b,f). Against the 51 speleothem records the emulator reaches a correlation of 0.57 with an RMSE of 0.92 ‰, against MPI-ESM-wiso's 0.63 and 0.79 ‰; against the 14 ice cores the emulator reaches 0.45 and the physical model 0.28. An emulator that never saw an isotope model of these nine climates recovers the observed mid-Holocene change about as well as a model that resolves the isotopes explicitly.

The framing choice is visible in this comparison too, and against real data it is a sharper test than any model-to-model diagnostic (Fig. S5). Holding the learner set fixed at the three tree ensembles and changing only the framing, the speleothem correlation falls from 0.57 to 0.43 for PMIP4 and from 0.67 to 0.49 for MPI-ESM, with the ice cores following the same direction. The trees are the family that survives the raw framing in the model-to-model comparison, so this residual penalty is attributable to the framing itself rather than to weak learners, and it appears for both application targets. Against the same sites AWIESM2-wiso, the model the emulator was trained on, reaches 0.61 on the speleothems and 0.18 on the ice cores, so the emulated fields under the anomaly framing are not merely close to a physical isotope model but in places ahead of one.

These numbers should be read with the archives in mind. Speleothem $\delta^{18}O$ is calcite on the VPDB scale and ice-core $\delta^{18}O$ is snow, while the emulator predicts precipitation $\delta^{18}O$ on the VSMOW scale, and no calcite or firn transfer function is applied. The comparison therefore tests the sign and the spatial pattern of the change rather than its absolute amplitude. Much of the constant offset cancels in a mid-Holocene minus pre-industrial difference, which is why the difference is the defensible quantity to compare and why the modern comparison, where no such cancellation occurs, carries the larger caveat.

2.7 Isotope–climate relations in the emulated fields

A final test asks what the emulated fields contain physically. Classical relations between $\delta^{18}O$ and climate are diagnosed in the same way from the two models that simulate isotopes explicitly and from the nine emulated PMIP4 models, in six latitude bands, both as a spatial gradient across grid points and as a temporal slope between the mid-Holocene and the pre-industrial state (Fig. 9). Agreement between the two groups indicates that the relation belongs to the simulated climate rather than to a prior the emulator carried over from its training model.

The spatial temperature effect passes this test. The emulated models reproduce the observed latitude dependence without being told about it, with slopes near 0.6 ‰ °C^{-1} at high northern latitudes, between 0.55 and 0.70 ‰ °C^{-1} at high southern latitudes, and a collapse to about 0.1 ‰ °C^{-1} in the tropics, all within the range spanned by AWIESM2-wiso and MPI-ESM-wiso. The spatial amount effect behaves the same way, near -1.1 ‰ per unit of log precipitation in the northern mid-latitudes for the emulated models against -1.06 ‰ in the truth. The elevation effect is reproduced as closely, near -0.9 ‰ per kilometre in the northern mid-latitudes and between -1.2 and -1.6 ‰ per kilometre in the tropics, steepening to -5.6 and -6.9 ‰ per kilometre over the two ice sheets, and the emulated models sit within the range of the two isotope-enabled ones in every band. That the emulator recovers three independent classical relations, each with its own latitude structure, is the strongest evidence that the fields carry physics rather than a memorised pattern.

The temporal slopes tell a more cautionary story. They run systematically below the spatial ones, with a median ratio of 0.49 over the 66 combinations of source and band, and the shortfall is far from uniform. The ratio is 0.65 between 60 and 90° N and 0.47 between 60 and 90° S, but only 0.31 between 30 and 60° N and 0.11 between 30 and 60° S. A palaeothermometer calibrated on the modern spatial gradient would therefore understate a mid-Holocene temperature change by a factor of two at high latitudes and by much more in the mid-latitudes. The emulator also reproduces the temporal slopes less faithfully than the spatial ones. The scatter of the emulated models around the truth is wider in the temporal direction, and between 0 and 30° S the emulated temperature slope even changes sign relative to the truth. The emulated fields are consequently reliable for present-day gradients and large-scale change patterns, and should not be used to calibrate a band-by-band temperature reconstruction.

3 Discussion

The emulator transfers between isotope-enabled models with a skill that has not been demonstrated before. Wider et al. (2023) found that a neural network trained on one model lost most of its skill when applied to another, falling to the level of pixel-wise linear regression. The emulator developed here keeps a land-mean temporal correlation of 0.66 and reproduces the mid-Holocene change at a spatial correlation of 0.87 on a model it never saw. The results here reproduce that earlier failure and also explain it, since the neural network and the two kernel methods behave in exactly the reported way while the tree ensembles do not. Two differences separate the two outcomes. The emulator learns from physical fields at single grid points instead of gridded images, so it never acquires the model-specific spatial structure that a convolutional network must learn. Geographic coordinates are also excluded from the predictors, so the emulator cannot fall back on position as a proxy for the field it is meant to predict. That this exclusion costs nothing physical is visible in the predictor importances, which recover the regional division between the temperature effect and the amount effect without ever being given a coordinate. The emulator can therefore stand in for an explicit isotope scheme when a model does not carry one, provided the learner is chosen with transfer in mind.

The comparison with the geological record supports that reading from outside the model world altogether. Against 51 speleothem records the emulated mid-Holocene change reaches a correlation of 0.57 and against 14 ice cores 0.45, while the two models that resolve isotopes explicitly reach 0.63 and 0.28 at the same sites. An emulator that never saw an isotope simulation of these climates therefore recovers the observed change about as well as a model that carries the isotopes itself. That agreement is what licenses applying it to models with no isotope scheme at all.

The more general lesson concerns how skill is measured. A learner that fits the training model tightly can be worthless on a foreign one, and the ranking of the seven methods by internal validation error is close to the reverse of their ranking by cross-model skill. Any emulator intended for a model other than the one it was trained on should therefore be selected on an independent model rather than on a held-out split of its own training data. The same caution applies to hyperparameter tuning, which improved the internal score of two methods here without improving their transfer.

Removing the inter-model base state matters more than the choice of learner. In the raw framing the kernel methods lose the mid-Holocene signal completely, with correlations near zero, while the tree ensembles reach 0.84 to 0.87. Expressing the predictors as departures from the pre-industrial climatology of the target model lifts every method to between 0.85 and 0.92, kernel ridge included. What distinguishes the tree methods is thus not that they alone transfer, but that they are the only family that works in both framings, which makes them the safe default when a suitable baseline is unavailable. The anomaly framing has its own cost, since its output carries no absolute level and cannot deliver an absolute $\delta^{18}O$ field.

The attribution of uncertainty is correspondingly clearer than a raw comparison would suggest. Across the nine PMIP4 models the spread introduced by the choice of climate model is about three times that introduced by the choice of learning method. An earlier comparison based on the raw framing gave a much larger method spread, but that number is dominated by the two kernel methods that collapse under this framing and it overstates the real ambiguity of the product. Reporting an ensemble of learners remains worthwhile, since it exposes such a collapse where a single method would hide it, and the emulated change should still be read as a model-driven quantity rather than a method-driven one.

The absolute and the relative use of the emulator fail in different ways. Every method reproduces the spatial pattern of the pre-industrial field at a correlation of at least 0.97, yet the root-mean-square error on an independent model ranges from 1.2 ‰ for the random forest to 6.4 ‰ for the linear baseline. An emulated absolute field should consequently carry a stated uncertainty of one to two per mil even where its pattern looks perfect, while differences between two climate states are the more robust product. This distinction matters for proxy comparison, where absolute $\delta^{18}O$ and the change between periods are used for different purposes.

The isotope-climate relations diagnosed from the emulated fields support a physical reading of what the emulator learned, and at the same time set a limit on its use. The spatial temperature effect and its latitude dependence emerge in the nine emulated models without being prescribed, and they agree with the two models that simulate isotopes explicitly. The temporal slopes between the mid-Holocene and the pre-industrial state fall well below the spatial ones, by a median factor of two and by considerably more in the mid-latitudes. A palaeothermometer calibrated on modern spatial gradients would therefore understate past temperature change, a discrepancy already documented for ice cores (Jouzel et al., 1997) and now visible across an ensemble of models. The emulator reproduces these temporal slopes less faithfully than the spatial ones, so the emulated fields are best used for spatial patterns and large-scale change and not for a band-by-band temperature calibration.

Several limitations frame the interpretation. The emulator is trained on a single model, so its errors inherit the structural biases of that model, and the gap of about 0.1 in temporal correlation between the self-reconstruction and the cross-model test is the visible cost of this choice. Skill is lowest in arid regions, where precipitation is scarce and the precipitation-weighted $\delta^{18}O$ is noisy. The amount effect diagnosed in the polar bands is not a true amount effect, since polar precipitation and temperature are strongly collinear there and the regression attributes part of the temperature control to precipitation. The anomaly path requires the baseline of the target period to be taken from the target model itself, and a mismatched baseline collapses the mid-Holocene change signal. The comparison against real archives carries its own caveat: speleothem $\delta^{18}O$ is calcite on the VPDB scale and ice-core $\delta^{18}O$ is snow, while the emulator predicts precipitation $\delta^{18}O$ on VSMOW, and no transfer function is applied between them. Much of that constant offset cancels in a mid-Holocene minus pre-industrial difference, which is why the difference rather than the absolute level is the quantity compared here, and why the modern comparison carries the heavier caveat of the two.

The approach extends naturally beyond the mid-Holocene. Training on additional isotope-enabled simulations would reduce the single-model dependence, and applying the emulator to colder periods such as the Last Glacial Maximum would test its behaviour outside the range of the training climate. A learned emulator that supplies consistent $\delta^{18}O$ for the full PMIP model ensemble would allow the many isotope-free models to be brought into direct comparison with the proxy record.

4 Methods

Training data. The emulator is trained on the isotope-enabled transient simulation eh2pi2.5, carried out with AWIESM2-wiso. The atmosphere component is ECHAM6-wiso, built on the general circulation model ECHAM6 (Stevens et al., 2013) and enhanced by the three stable water isotope tracers $H_2{}^{16}O$, $H_2{}^{18}O$ and HDO (Cauquoin and Werner, 2021; Shi et al., 2023). The ice-ocean component is FESOM2 (Danilov et al., 2017; Sidorenko et al., 2019). The atmospheric grid is T63L47, about 1.9° horizontal resolution with 47 vertical levels. The transient spans 8199 model years, with the first year at 8.2 ka BP and the last near 1950 CE. Monthly precipitation $\delta^{18}O$ is the prediction target. The predictors are fourteen climate fields, namely temperature, specific humidity and geopotential height at 850 and 500 hPa, the zonal wind at both levels and the meridional wind at 850 hPa, together with surface temperature, precipitation, vertically integrated cloud water, cloud cover and outgoing longwave radiation. Surface elevation and top-of-atmosphere insolation complete the predictor set. Latitude and longitude are deliberately excluded, so that the emulator cannot resolve a grid point by its position.

Validation and application data. Cross-model validation uses MPI-ESM-wiso (Cauquoin et al., 2019), which provides its own precipitation $\delta^{18}O$ for a mid-Holocene and a pre-industrial state. The mid-Holocene simulation contains an initial adjustment of roughly 150 years, so only its last 350 years are used, while the pre-industrial control is used in full at 353 years. The emulator is applied to nine PMIP4 mid-Holocene simulations, selected for having the required predictor fields, together with their pre-industrial controls (Otto-Bliesner et al., 2017; Brierley et al., 2020), each taken as a 100-year slice. All fields are regridded to the T63 grid of the training model.

Emulator and training. Seven regression methods from four learner families are trained through a common interface. The tree family comprises gradient boosting (Chen and Guestrin, 2016), a random forest (Breiman, 2001) and histogram gradient boosting. The kernel family comprises kernel ridge regression and support vector regression, both with radial basis functions and both fitted on a random subsample of at most fifteen thousand rows per region to keep the kernel computation tractable. A feed-forward neural network with two hidden layers of 128 and 64 units and a ridge regression complete the set. Separate models are fitted for each calendar month and for precipitation-based sub-regions of each continent, with the two ice sheets treated as single regions. One model year in eight enters the training set, which gives 1025 of the 8199 years, and the validation split is taken by whole years rather than by grid points so that no year contributes to both training and validation. Each training sample is weighted by its monthly precipitation, which aligns the loss with the precipitation-weighted product and down-weights the noisy dry-season samples. Where the surface lies above 2000 m the 850 hPa fields are treated as missing, since that level is below ground. The gradient-boosting hyperparameters are tuned with a time-block cross validation that maximises the held-out temporal correlation, and the tuned values are adopted only after they are confirmed to improve the independent cross-model skill.

Preprocessing. Two preprocessing paths are used. The raw path predicts absolute $\delta^{18}O$ from absolute predictor fields. The anomaly path expresses both the predictors and the target as departures from a climatology, taken during training as the mean of the last thousand years of the transient. When the emulator is applied to another model the baseline is recomputed from that model's own pre-industrial state, which aligns the two climates before the learned mapping is applied. A mismatched baseline removes the mid-Holocene signal, so this step is the critical one in the anomaly path. Insolation is retained in all configurations, because removing it flattens the orbital-scale trend of the transient.

Evaluation. Skill is evaluated on the precipitation-weighted annual $\delta^{18}O$ over land, as a per-grid-point temporal correlation of the annual series and as the spatial correlation and root-mean-square error of the multi-year climatology. Grid points with annual precipitation below 50 mm are excluded, since the precipitation-weighted isotope value is not meaningful where there is almost no precipitation, and the two ice sheets are exempt from this filter. Emulated values beyond 60 ‰ in magnitude are discarded as unphysical, which affects only the linear and neural methods at a small number of out-of-range grid points. Skill on the training model is computed after the 1025 training years are removed from the time axis, leaving 7174 unseen years. For the anomaly path both the emulated and the simulated series are referred to the same climatology before any comparison, since the anomaly output lives in the reference frame of the training model.

Proxy data. Three archives are used. Modern precipitation $\delta^{18}O$ comes from 231 GNIP stations as a precipitation-weighted annual mean, restricted to the 70 stations with more than four years of record because a one- or two-year mean is a noisy estimate of a climatology. Speleothem $\delta^{18}O$ comes from 57 SISALv2 entities (Comas-Bru et al., 2020) with a pre-industrial and a mid-Holocene mean per entity, which are 40 distinct caves, so entity-level statistics over-weight the repeated sites and site-averaged values are reported alongside. Ice-core $\delta^{18}O$ comes from 15 cores for the pre-industrial state and 14 for the mid-Holocene minus pre-industrial change. Sites are matched to the model grid by bilinear interpolation over land; a site whose cell is ocean in the model is snapped to the nearest land cell within 3°, and sites with no land cell in that radius are excluded, which removes 30 GNIP stations and 6 caves. Statistics are reported per archive rather than pooled, because a single correlation over archives spanning −57 to −1 ‰ is dominated by the offset between polar ice and tropical caves. No calcite or firn transfer function is applied, so speleothem values remain calcite on the VPDB scale and ice-core values remain snow while the emulator predicts precipitation $\delta^{18}O$ on VSMOW.

Isotope-climate relations. Slopes of $\delta^{18}O$ against surface temperature and against the logarithm of precipitation are computed in six latitude bands. The spatial slope is a regression across land grid points within a band for the pre-industrial state, and the temporal slope is the ratio of the mid-Holocene minus pre-industrial change in $\delta^{18}O$ to the corresponding change in temperature or log precipitation. The two isotope-enabled models and the nine emulated PMIP4 models are treated identically. AWIESM2-wiso has no separate mid-Holocene experiment, so a 1000-year window centred on 6 ka BP is taken from the transient and matched in length to the pre-industrial window formed by its last thousand years, with training years removed from both.

Data and code availability

The emulator code and the reconstruction outputs are available in the project repository. The training simulation and the MPI-ESM-wiso validation data are available from the respective model archives.

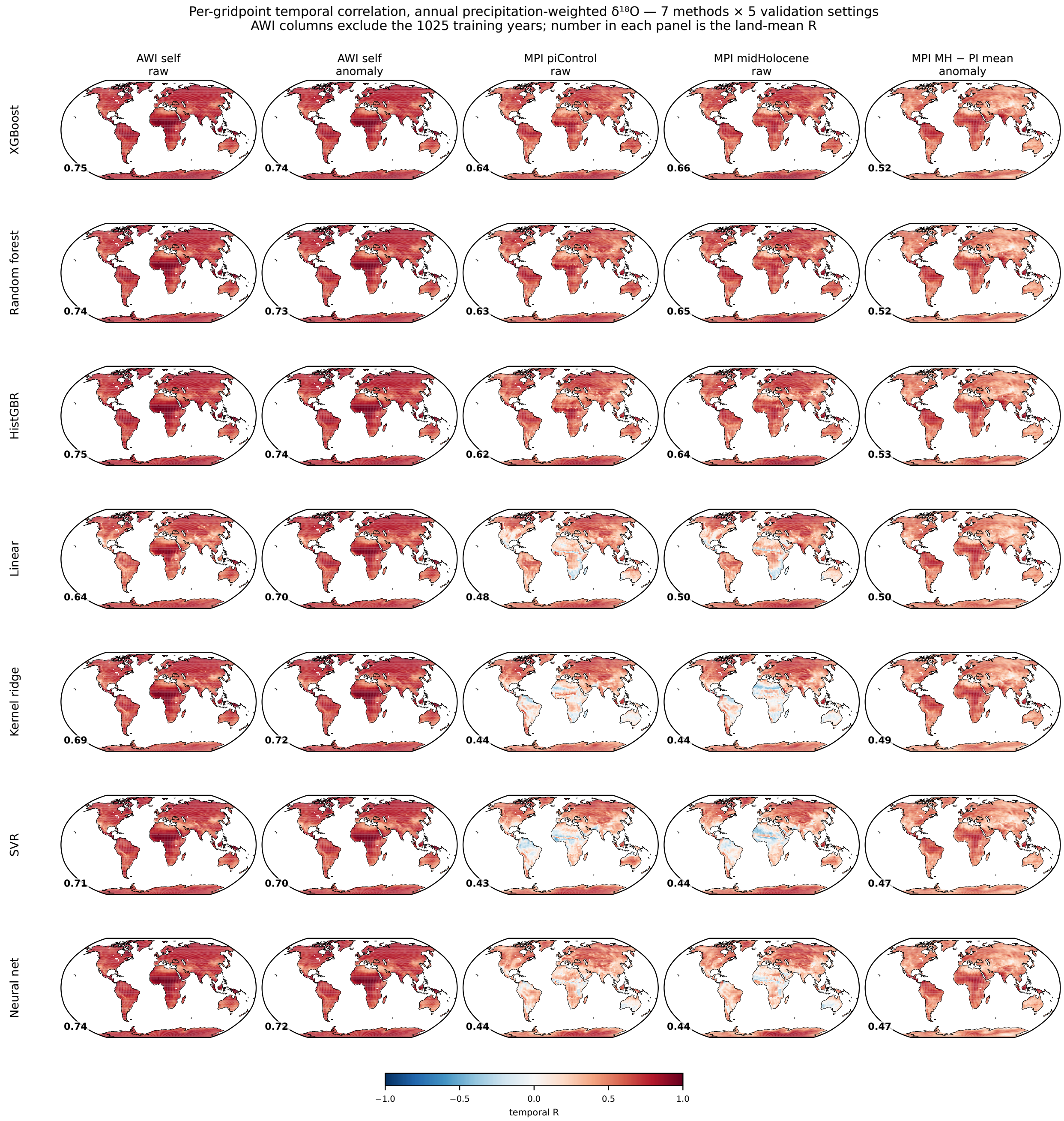

Figure 1: Per-grid-point temporal correlation between the emulated and the simulated annual precipitation-weighted $\delta^{18}O$, for seven learners (rows) and five validation settings (columns). The columns are the AWIESM2-wiso self-reconstruction under the raw and the anomaly framing, the MPI-ESM-wiso piControl and mid-Holocene states under the raw framing, and the MPI-ESM-wiso mid-Holocene departure from the piControl mean state under the anomaly framing. The two AWIESM2-wiso columns exclude the 1025 training years, leaving 7174 unseen years. The number in each panel is the land-mean correlation. Reading along a row gives the cost of leaving the training model; reading down a column gives the spread between learners, which is small within the training model and large across models.

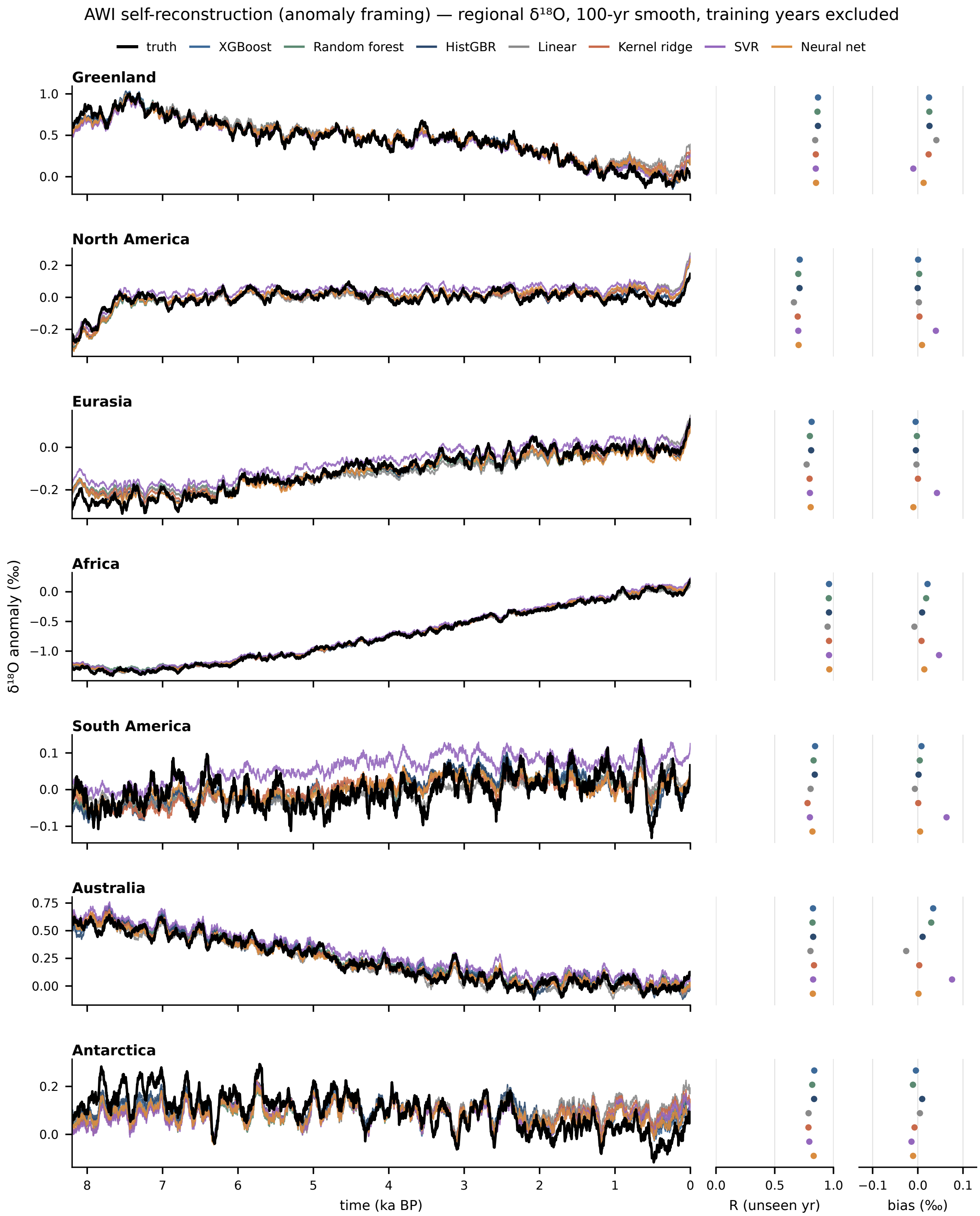

Figure 2: Regional precipitation-weighted δ^{18}O of the AWIESM2-wiso self-reconstruction under the anomaly framing, for seven regions ordered north to south. Black is the simulated series and the coloured lines are the seven learners, all smoothed with a 100-year running mean. Training years are masked before smoothing and excluded from the statistics. The strips on the right give the per-region correlation and the mean bias for each learner. Both sides are referenced to the same fixed late-Holocene climatology, so the bias strip reports a real offset rather than being forced to zero.

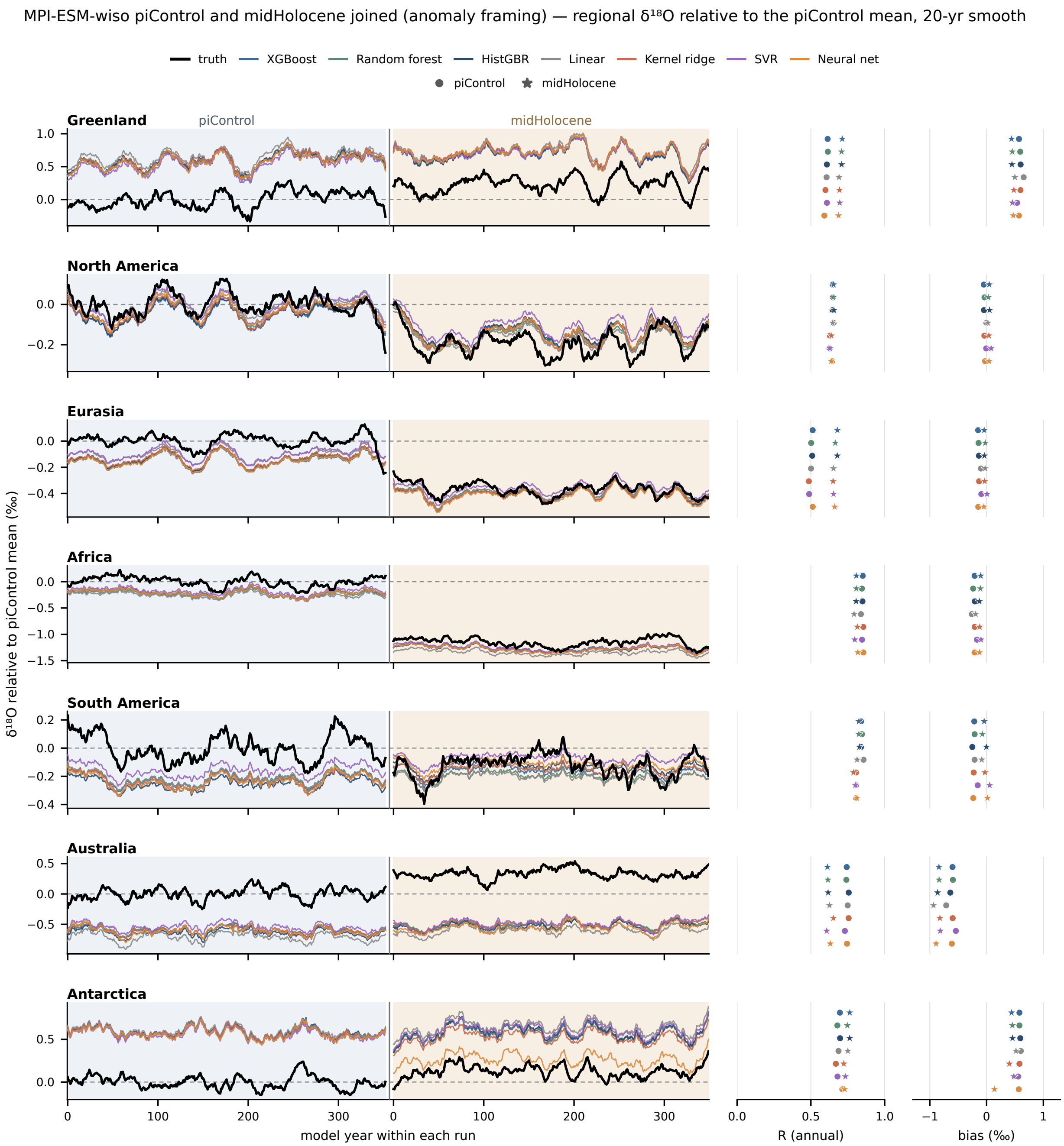

Figure 3: MPI-ESM-wiso piControl and mid-Holocene drawn as one continuous series per region, with δ^{18}O expressed relative to the piControl mean state so that the piControl segment sits near zero by construction and the mid-Holocene segment is a directly readable departure. Shaded blocks separate the two states, which are independent simulations rather than one time axis, and the model year restarts at zero for the mid-Holocene block. Black is the simulation and the coloured lines are the seven learners. The strips on the right give the per-region correlation and mean bias, with circles for the piControl and stars for the mid-Holocene.

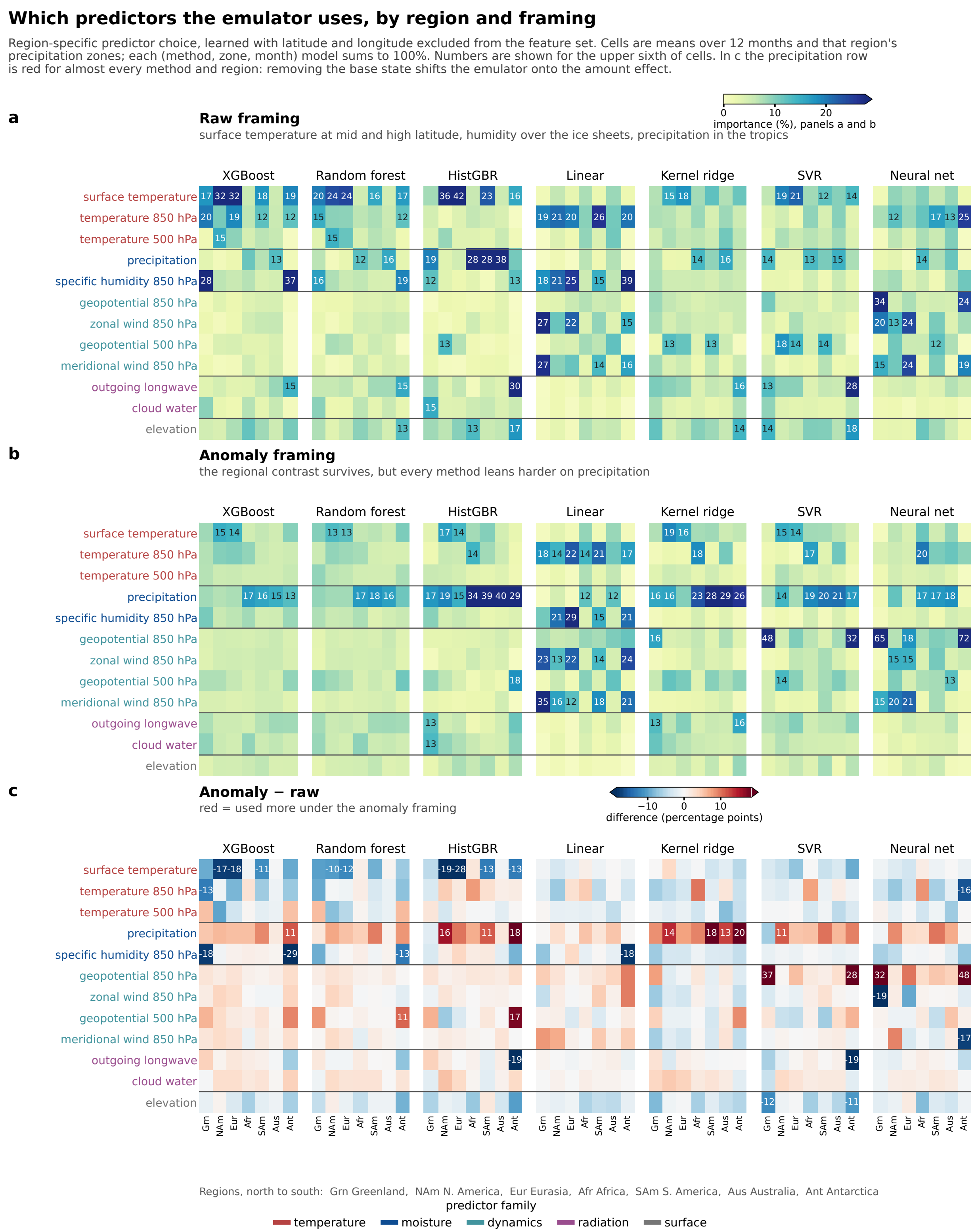

Figure 4: Predictor importance of the emulator by learner, region and preprocessing framing, with latitude and longitude excluded from the predictor set. (a) Raw framing. (b) Anomaly framing, on the same colour scale. (c) The difference between them in percentage points. Cells are means over the twelve months and over the precipitation zones of each region, and each individual model sums to 100 %. Predictors are ordered by mean importance and grouped by physical family, colour-coded in the axis labels. The tree ensembles use surface temperature across the mid-latitude continents, boundary-layer humidity over the two ice sheets and precipitation over Africa and Australia, which is the textbook division between the temperature effect and the amount effect. Panel (c) shows that removing the base state shifts every learner towards precipitation.

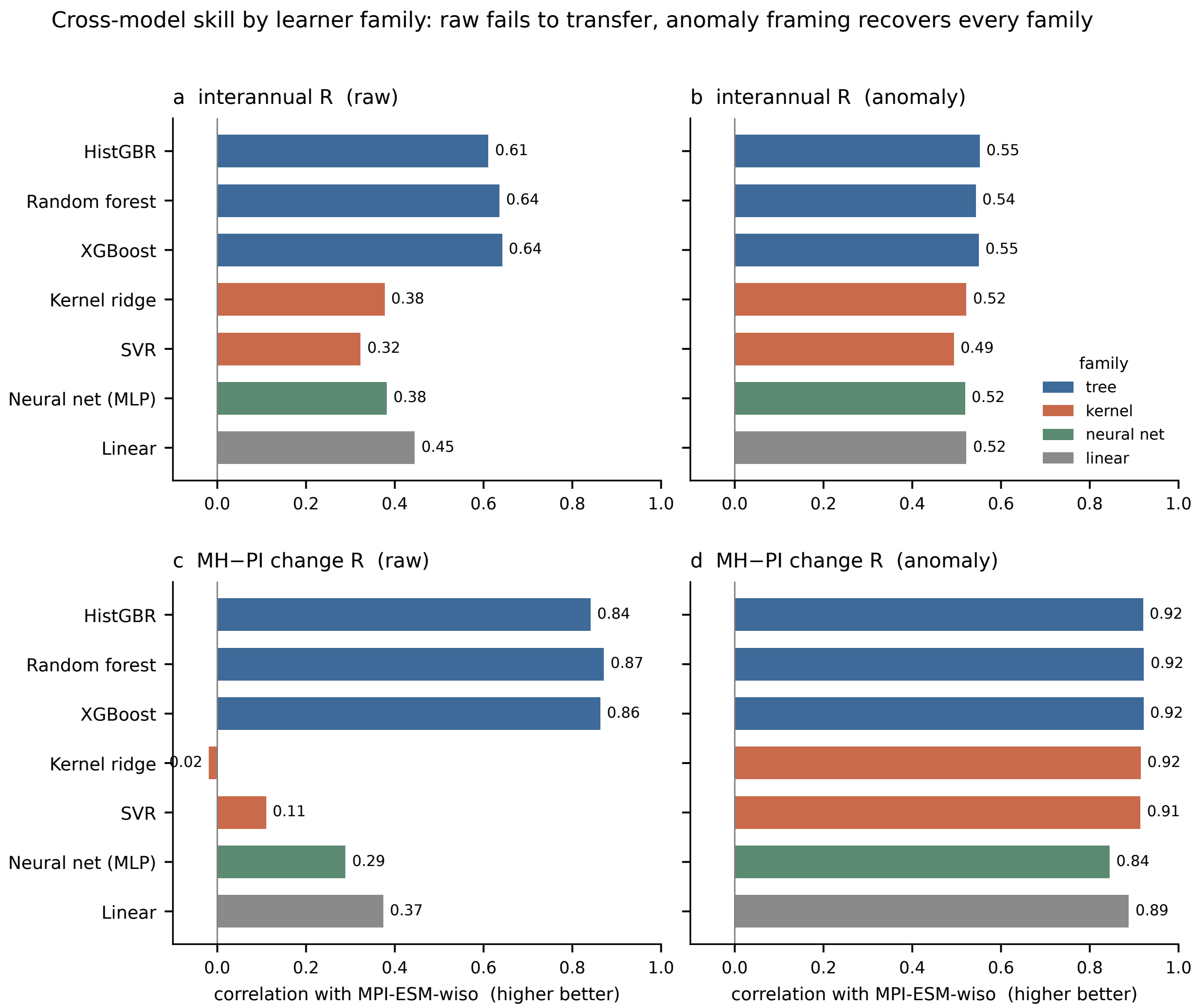

Figure 5: Cross-model skill of seven learners from four families: gradient boosting, random forest and histogram gradient boosting (tree), kernel ridge and support vector regression (kernel), a feed-forward neural network, and ridge regression (linear). (a, b) Precipitation-weighted temporal correlation on MPI-ESM-wiso for the raw and the anomaly framing. (c, d) As in (a)–(b), but for the spatial correlation of the mid-Holocene minus pre-industrial change. The ranking of the learners inside the training model does not survive the transfer, and the anomaly framing closes most of the gap between the families.

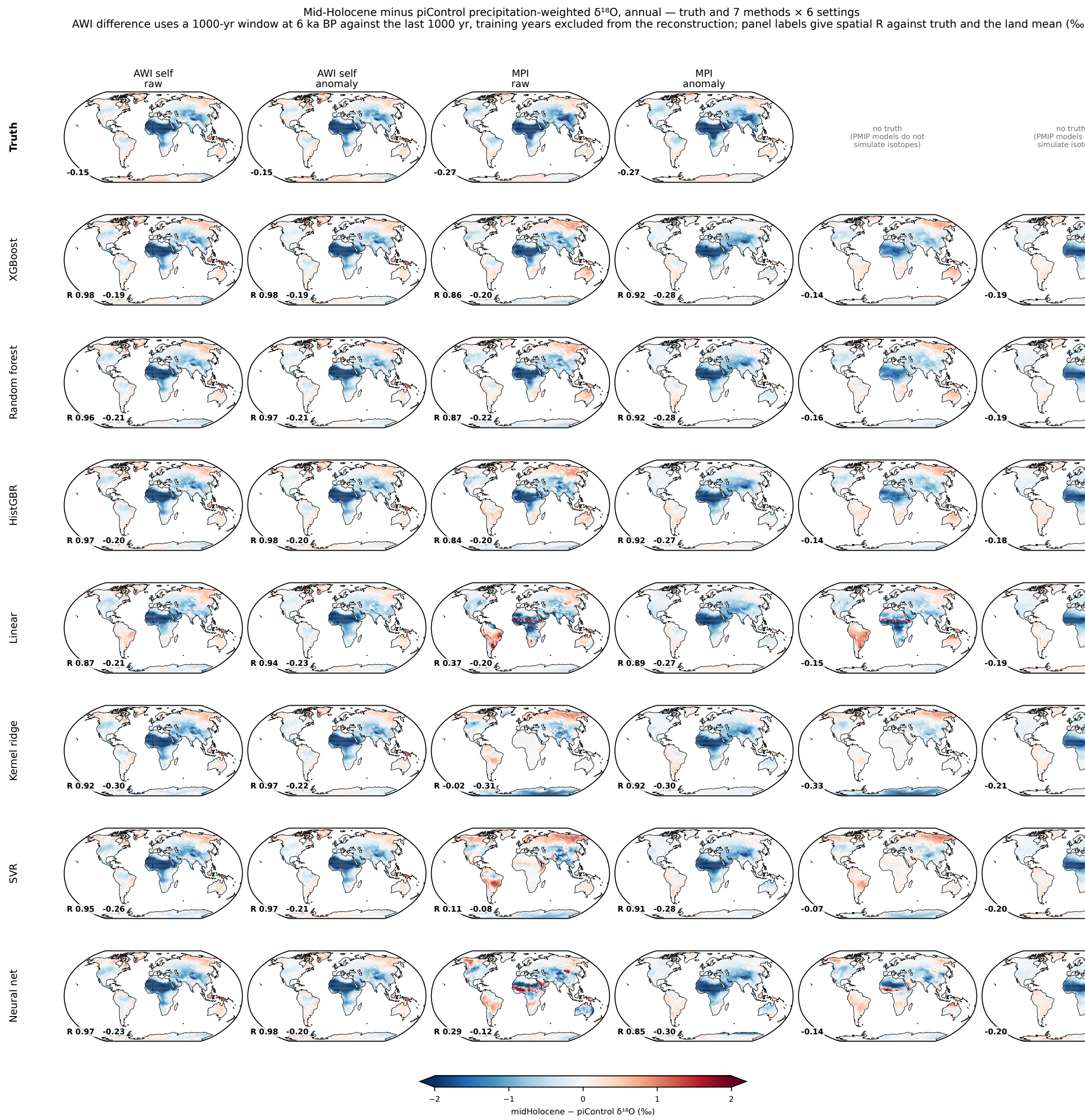

Figure 6: Mid-Holocene minus pre-industrial precipitation-weighted δ^{18}O, for the simulated field and seven learners (rows) against six settings (columns): the AWIESM2-wiso self-reconstruction, MPI-ESM-wiso and the nine-model PMIP4 ensemble mean, each under the raw and the anomaly framing. The truth row is empty for the two PMIP4 columns because those models do not simulate water isotopes, which is the reason the emulator is applied to them. Every column forms the difference by subtracting an explicitly reconstructed pre-industrial field. For AWIESM2-wiso the two states are a 1000-year window centred on 6 ka BP and the last 1000 years of the transient, and the reconstruction uses only years excluded from training. Panel labels give the spatial correlation against the truth of the same column and the land mean. Under the raw framing the kernel and linear learners lose the Green Sahara depletion entirely; under the anomaly framing all seven recover it.

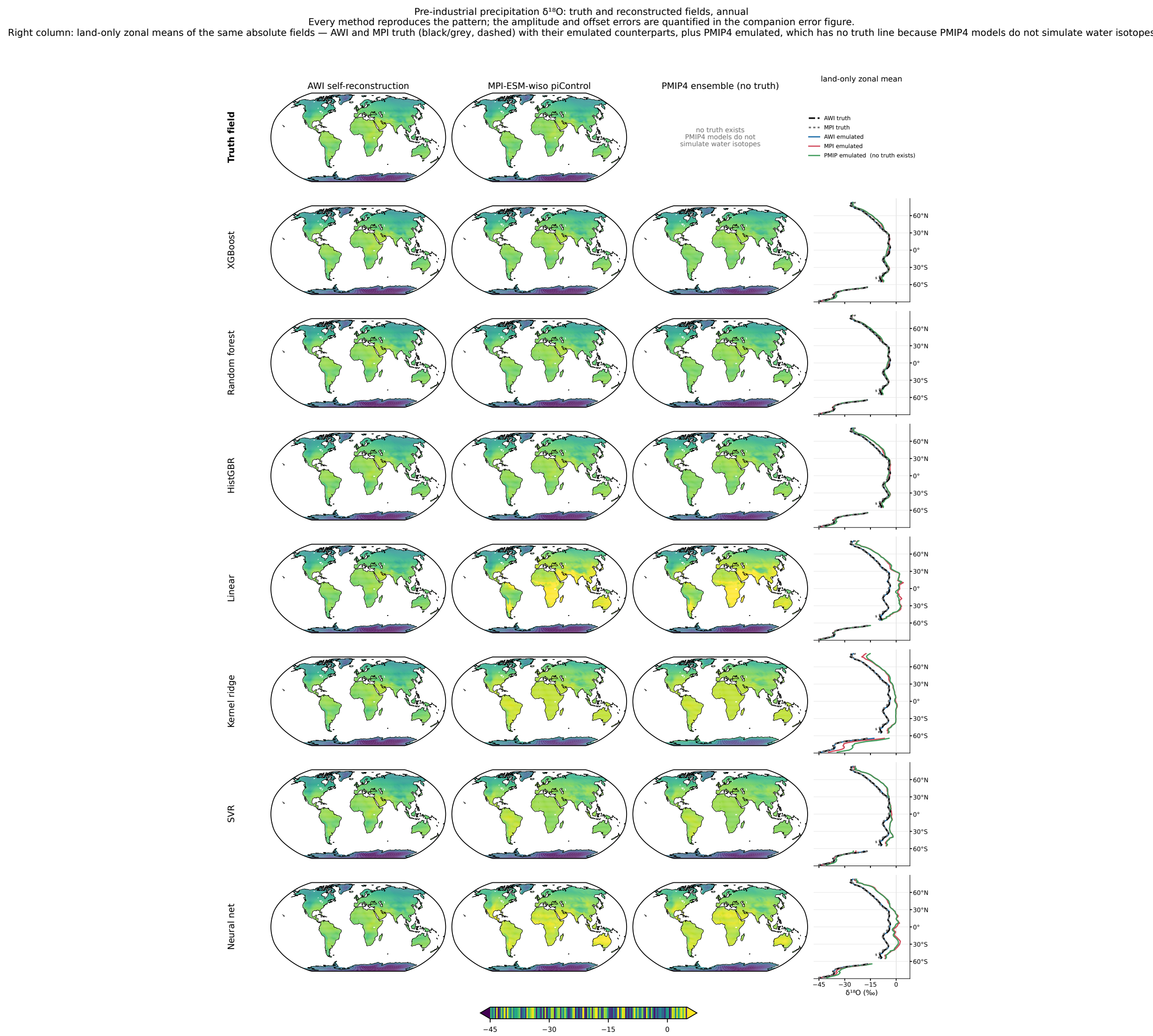

Figure 7: Pre-industrial precipitation-weighted δ^{18}O as an absolute field, for the simulated field and seven learners (rows) against three settings (columns): the AWIESM2-wiso self-reconstruction, MPI-ESM-wiso piControl, and the nine-model PMIP4 ensemble mean, which has no simulated counterpart. All panels share one absolute colour scale. The rightmost column gives the land-only zonal mean of the same fields, with the two simulated profiles dashed in black and grey and the emulated profiles in colour. Every learner reproduces the spatial pattern, with spatial correlations of 0.97 or better, so the panels are difficult to separate by eye; the error is in the level, and it is quantified per learner in the companion analysis rather than shown here.

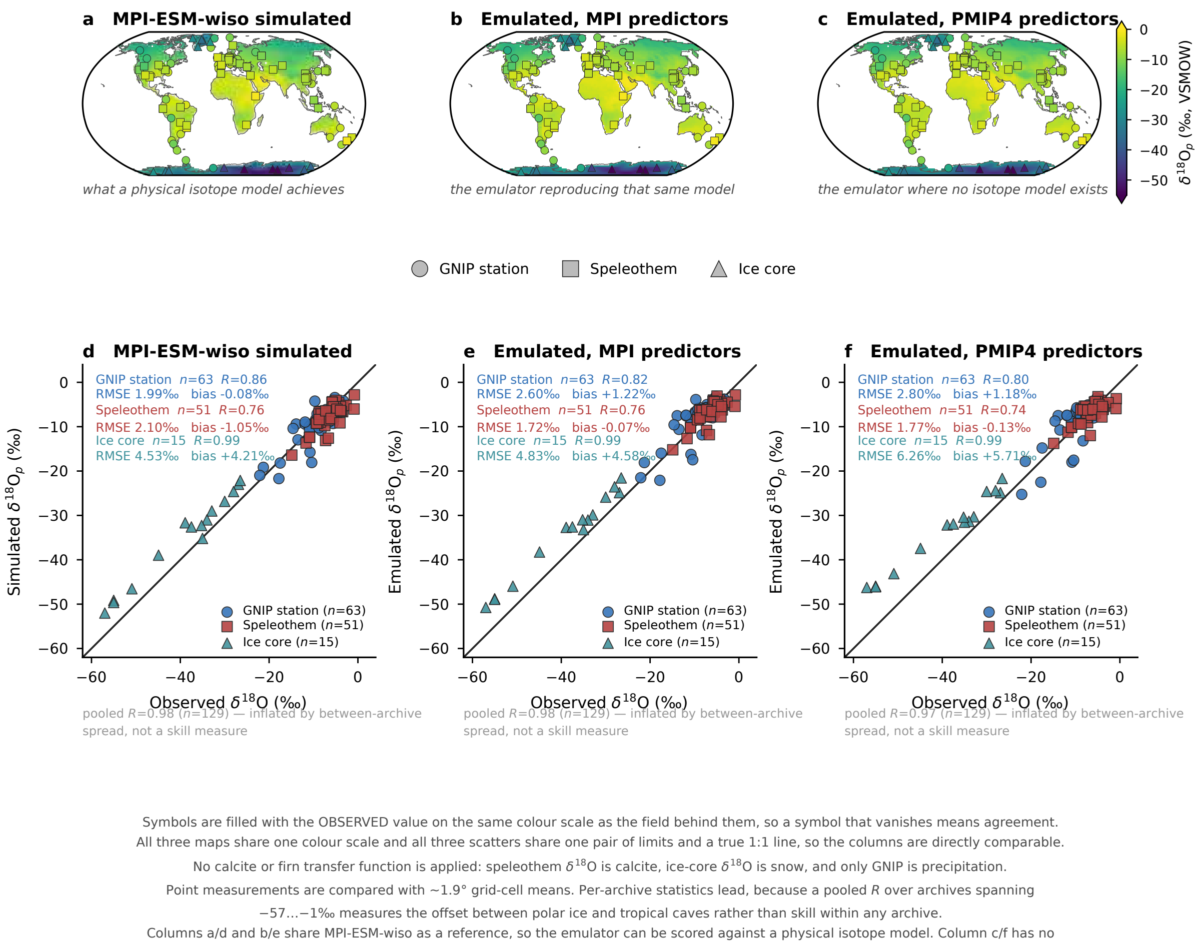

Figure 8: Emulated pre-industrial precipitation δ^{18}O against observations at 129 sites from three archives: 63 GNIP stations with records longer than four years, 51 SISALv2 speleothem entities and 15 ice cores. (a–c) Maps of the field with the sites overlaid as symbols filled with the observed value on the same colour scale, so a symbol that vanishes into its background means model and observation agree. (d–f) The same three fields sampled at the sites, on shared axes with a 1:1 line. The columns are the simulated MPI-ESM-wiso field, the emulator driven by MPI-ESM predictors, and the emulator driven by PMIP4 predictors, that is, what a physical isotope model achieves, the emulator reproducing that same model, and the emulator where no isotope model exists. Statistics are given per archive because a single correlation over archives spanning -57 to -1 ‰ would mostly measure the offset between polar ice and tropical caves; the pooled value is greyed for that reason. Speleothem δ^{18}O is calcite on the VPDB scale and ice-core δ^{18}O is snow, whereas the emulator predicts precipitation δ^{18}O on VSMOW, and no transfer function is applied.

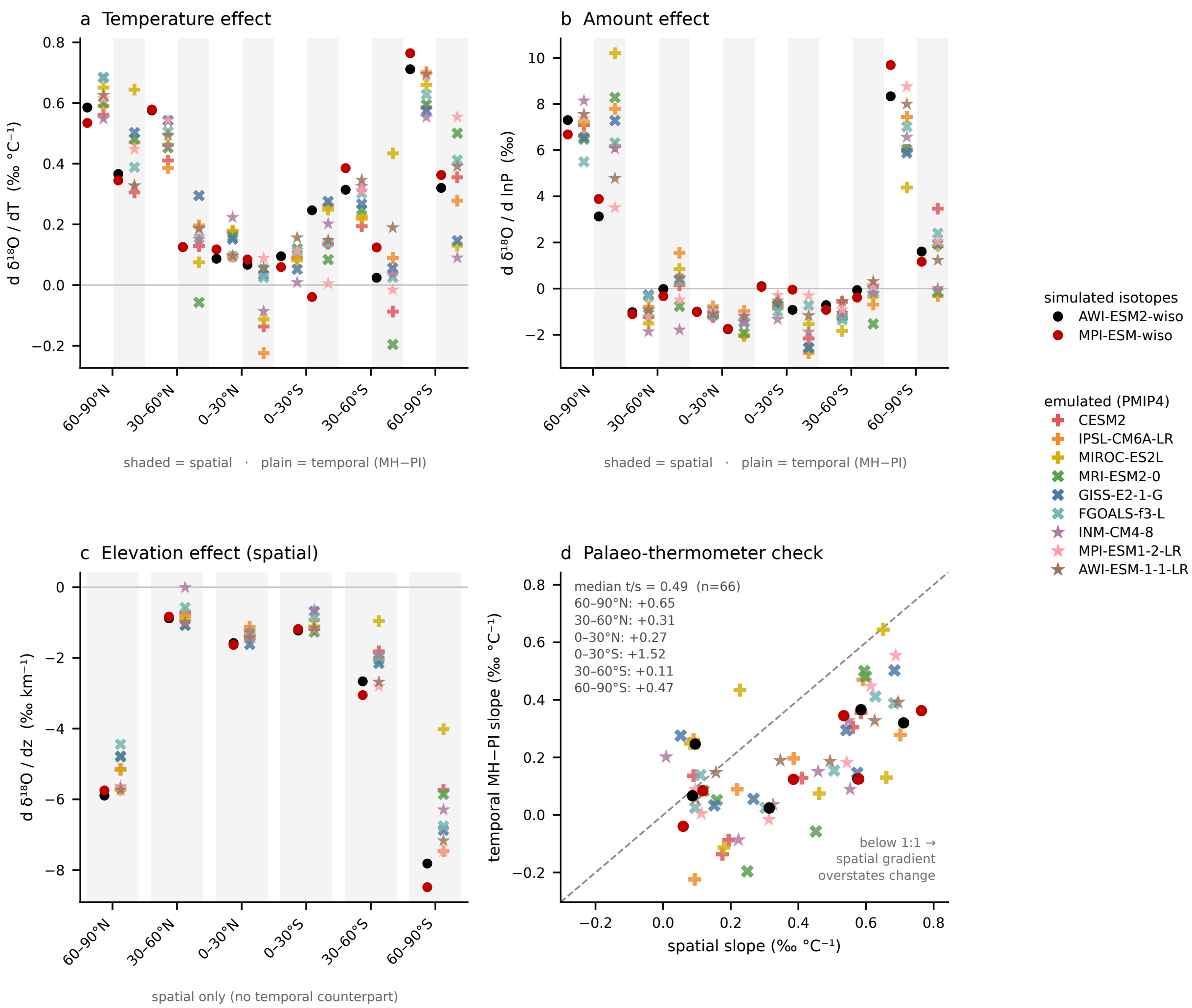

Figure 9: Isotope–climate relations by latitude band, diagnosed identically from the two models that simulate isotopes explicitly and from the nine emulated PMIP4 models. (a) Slope of $\delta^{18}O$ against surface temperature. (b) Slope against the logarithm of precipitation. In (a) and (b) the shaded half of each band is the spatial gradient across grid points and the plain half is the temporal mid-Holocene minus pre-industrial slope. (c) Slope against elevation, spatial only, since elevation does not change between the two states. (d) Temporal against spatial temperature slope, one symbol per model and band, with the 1:1 line and the median ratio per band. Small filled circles are the isotope-enabled models, black for AWIESM2-wiso and dark red for MPI-ESM-wiso, and the coloured crosses and stars are the emulated models. Overlap between the two groups indicates that a relation belongs to the simulated climate rather than to a prior carried over from the training model.

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Supplementary figures

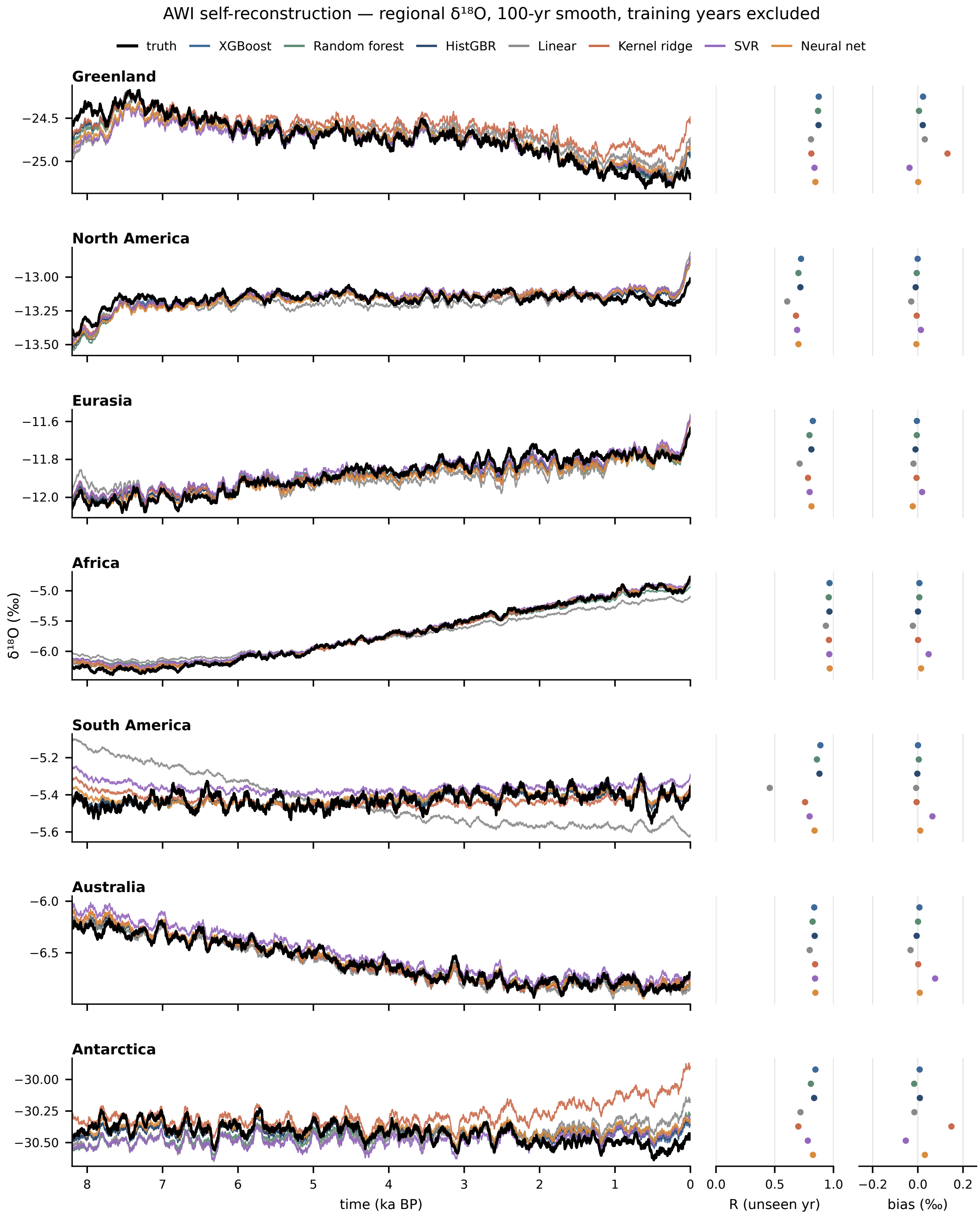

Figure S1: Regional precipitation-weighted $\delta^{18}O$ of the AWIESM2-wiso self-reconstruction under the raw framing, in absolute units. This is the only figure that keeps absolute levels, because a self-reconstruction has no inter-model offset to hide, and the orbital-scale trend and the regional level are the content. Otherwise as Fig. 2.

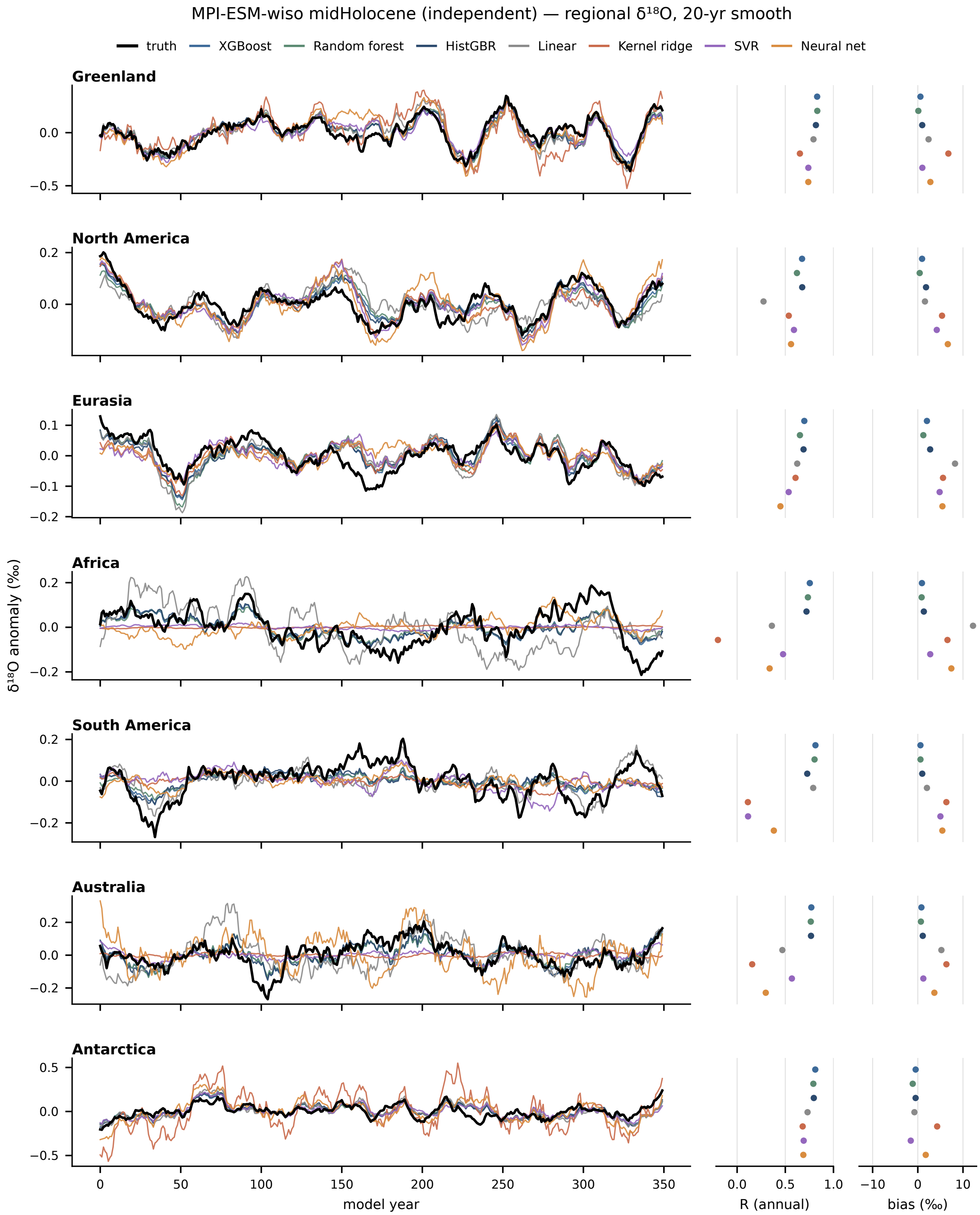

Figure S2: Regional precipitation-weighted δ^{18}O of the MPI-ESM-wiso mid-Holocene under the raw framing. Traces are de-meaned for display only, because the cross-model offset, up to 12.2 ‰ for the linear baseline over Africa, would otherwise flatten them into parallel lines; the offset itself is reported in the bias strip on the right. Otherwise as Fig. 2.

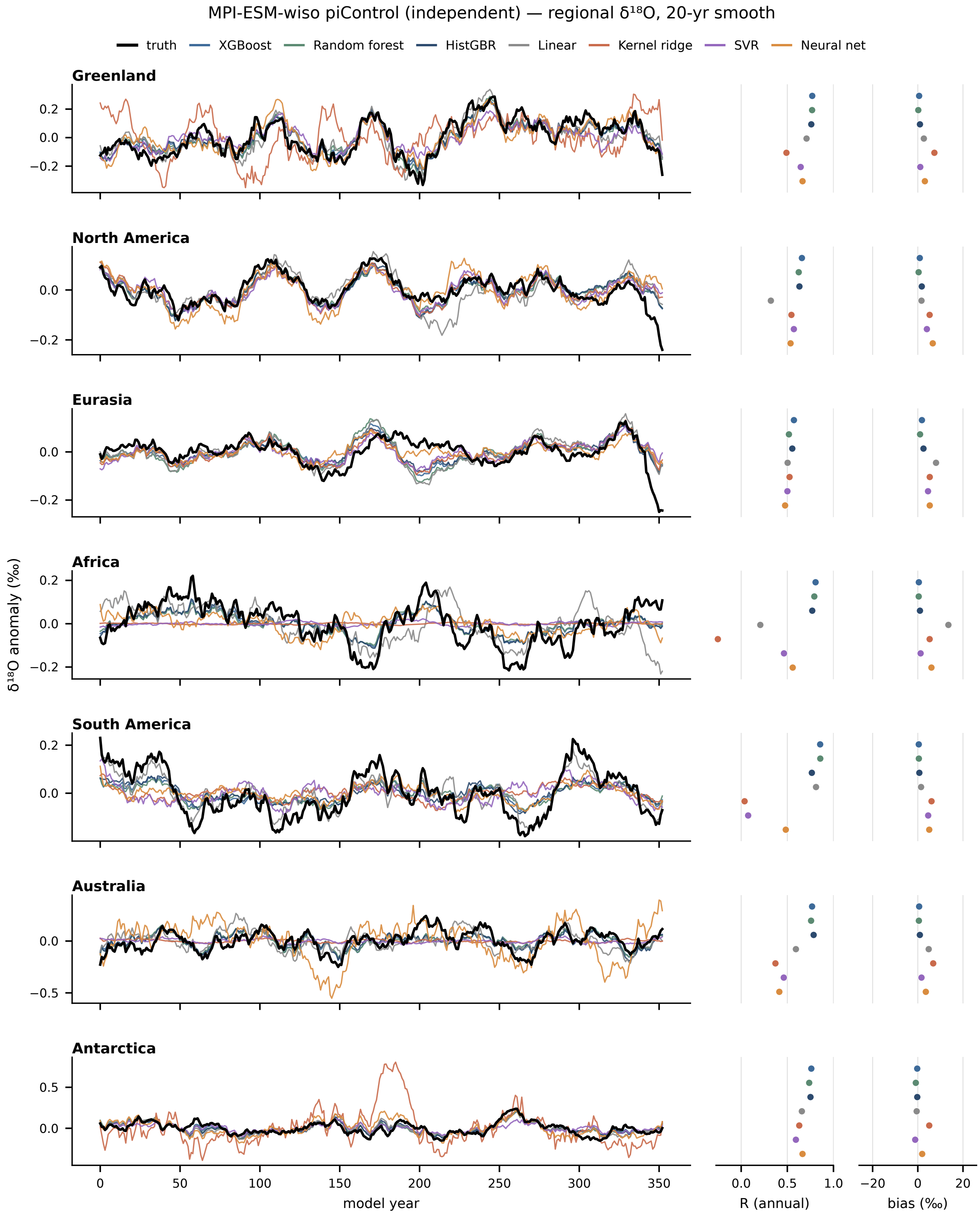

Figure S3: As in Fig. S2, but for the MPI-ESM-wiso piControl state. The same offsets appear in a second, independent base state, which is what identifies them as a difference in base state rather than a failure to track variability.

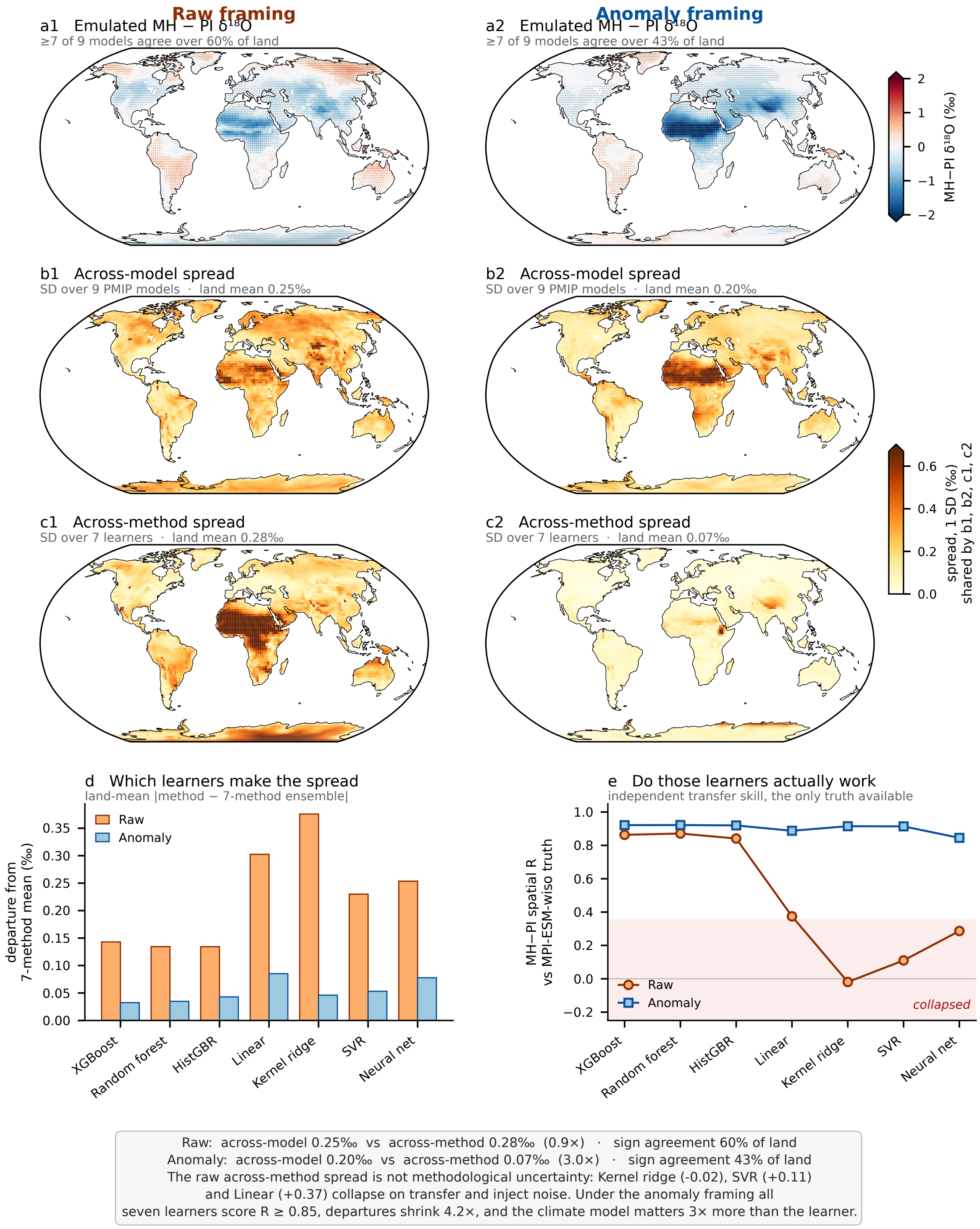

Figure S4: Raw against anomaly framing for the mid-Holocene minus pre-industrial δ^{18}O of the nine-model PMIP4 ensemble, evaluated against models rather than observations. The ensemble mean, the spread across the nine climate models and the spread across the seven learners are shown for both framings, with the two spread rows on one shared colour scale so that the dominant uncertainty is readable as area. Under the anomaly framing the climate model matters about three times more than the learner. Under the raw framing the apparent spread across learners is inflated by the kernel and linear methods that fail to transfer, which would misattribute a failure of two methods to genuine methodological uncertainty.

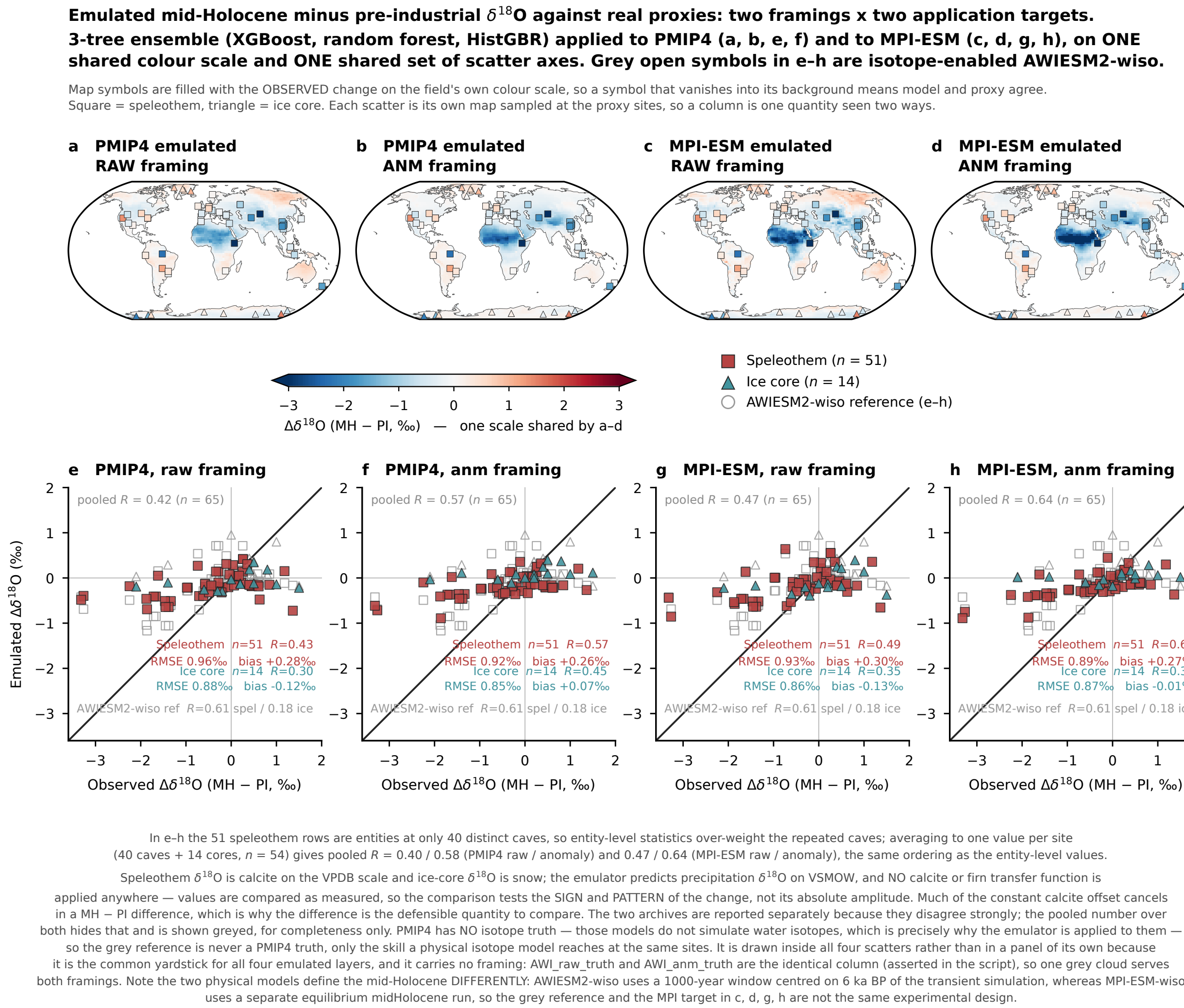

Figure S5: The same 65 proxy sites used to separate the effect of the preprocessing framing from the effect of the application target, with the ensemble held fixed at the three tree methods, which are the family that survives the raw framing. (a–d) Emulated mid-Holocene minus pre-industrial δ^{18}O on one shared colour scale, for PMIP4 under the raw and the anomaly framing and for MPI-ESM under the same two framings. (e–h) The field directly above each panel sampled at the proxy sites, on one shared set of axes with a 1:1 line, so a column is one quantity seen two ways. Reading across a pair varies the framing at fixed target; reading between pairs varies the target at fixed framing. Grey open symbols in every scatter are isotope-enabled AWIESM2-wiso at the same sites, drawn inside all four panels rather than in one of its own because it is the common yardstick and carries no framing. Note that the two physical models define the mid-Holocene differently: AWIESM2-wiso uses a 1000-year window centred on 6 ka BP of the transient simulation, whereas MPI-ESM-wiso uses a separate equilibrium simulation. Against real proxies the anomaly framing is better than the raw framing for both targets even with the tree methods, so the advantage established against models in Fig. 5 is not an artefact of the learners that fail to transfer. Speleothem δ^{18}O is calcite reported on the VPDB scale and ice-core δ^{18}O is snow, whereas the emulator predicts precipitation δ^{18}O on the VSMOW scale, and no calcite or firn transfer function and no scale conversion are applied, so the comparison tests the sign and the pattern of the change rather than its absolute amplitude.